Counting Problems Involving Symmetry: A Group Theoretic Perspective

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Overview

1. Three Motivating Problems
2. What is a Group?
3. Symmetry Groups
4. Group Actions
5. Burnside's Lemma
6. Solving the Three Problems
7. Conclusion

1 Three Motivating Problems

2 What is a Group?

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Three Questions to Start With

We begin with three counting problems that look simple but hide symmetry:

1. How many distinct dice can be constructed by labelling the 6 faces with numbers 1-6?
2. Three identical 52-card decks are mixed. How many distinct 3-card hands can be drawn?
3. How many distinct necklaces can be made using 2 black and 2 white beads?

These three problems motivate the need to understand overcounting caused by **symmetry**.

An Illustrative Game

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Why Symmetry Causes Overcounting

Example: Necklaces

- Rotating a necklace does not produce a new necklace.
- Reflecting a necklace also does not produce a new necklace.

Different drawings may represent the *same* physical object.

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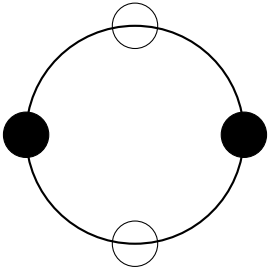

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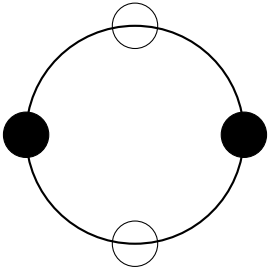

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Intuitive Idea of a Group

A **group** formalises the idea of symmetry.

Examples of "moves" that form groups:

- Rotating a cube in 3D space.
- Permuting (reordering) card positions.
- Rotating or flipping a necklace.

Common structure:

- You can **compose** moves.
- There is an **identity** move that does nothing.
- Every move can be **undone** (has an *inverse*).

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Formal Definition of a Group

A set G with a binary operation $\circ$ is a **group** if:

1. (Closure) If $g, h \in G$ then $g \circ h \in G$.
2. (Identity) There exists $e \in G$ such that $e \circ g = g$ and $g \circ e = g$.
3. (Inverses) For each $g \in G$ there exists $g^{-1} \in G$ with $g \circ g^{-1} = e = g^{-1} \circ g$.
4. (Associativity) $(g \circ h) \circ k = g \circ (h \circ k)$ for all $g, h, k \in G$.

In this lecture, our main examples of groups are:

- **Symmetric groups** S_n (all permutations of n positions).
- **Symmetry groups** of geometric objects: cube, square/necklace.

We only focus on **finite** groups (i.e., $|G| < \infty$).

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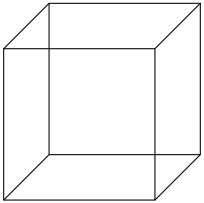
Rotational Symmetries of the Cube: the Symmetric Group S_4

The cube has 24 rotational symmetries[1].



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- These rotations form the **symmetric group** S_4.

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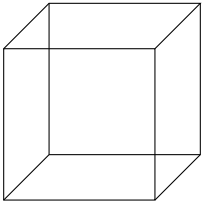

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Symmetries of 3 Positions: the Symmetric Group S_3

We label three positions by $1, 2, 3$.

$$1 \qquad 2 \qquad 3$$

- Any reordering (permutation) of the three positions is an element of S_3.
- Example: the cycle (123) sends $1 \mapsto 2$, $2 \mapsto 3$, $3 \mapsto 1$.

$$\begin{tikzcd} 1 \arrow[r, bend left] & 2 \arrow[r, bend left] & 3 \arrow[ll, bend left] \end{tikzcd}$$

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Symmetries of the Square: the Dihedral Group D_4

A square (or 4-bead necklace on a loop) has 8 symmetries:

- 4 rotations: $0°, 90°, 180°, 270°$.
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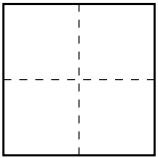

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Why Group Actions?

We want to let symmetries act on the sets we are counting.

- Cube rotations act on all possible labellings of faces.
- S_3 acts on ordered 3-card hands by permuting positions.
- D_4 acts on bead sequences placed at 4 positions around a square.

This leads to the formal notion of a **group action**.

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Definition: Group Action

A group G **acts** on a set X if each $g \in G$ gives a function $g : X \to X$ such that:

- Identity: $e \cdot x = x$ for all $x \in X$.
- Compatibility: $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$, $x \in X$.

Some key notions:

- **Orbit** of x : $Gx = \{g \cdot x \mid g \in G\}$[2].
- **Fixed point** of g : an x with $g \cdot x = x$.
- X^g : the set of all fixed points of g.
- G_x : the *group* containing all g fixing x.

[2] Note that orbits form an equivalence relation.

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Example: S_3 Acting on a 3-Card Hand

Let a hand be an ordered triple (a_1, a_2, a_3).

- The permutation (12) swaps the first two positions:

$$(12) \cdot (a_1, a_2, a_3) = (a_2, a_1, a_3).$$

- The permutation (123) cycles through the three positions:

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All hands reachable from one another by permutations lie in the same **orbit**.

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From Orbits to Counting

Our real goal: count *distinct* objects up to symmetry.

$$|X/G| = \text{number of orbits of } G \text{ acting on } X^3.$$

If the action has no non-identity element with fixed points (**"fixed-point free"**):

$$|X/G| = \frac{|X|}{|G|}.$$

But in many interesting problems, there *are* fixed points. We need a more general tool.

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$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|.$$

Interpretation:

- For each symmetry g, count how many objects look unchanged under g.
- Average this fixed-point count over the whole group.
- Then we get the number of orbits.

We will now use this to solve our three motivating problems.

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We start with $C = \{(g, x) \mid g \cdot x = x\}$:

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- $|C| = \sum_{x \in X} |G_x| = \sum_i \left(\sum_{x \in G \cdot x_i} |G_x| \right)$.
- For each orbit $G \cdot x_i$, $|G_x| = |G_{x_i}|$ for all $x \in G \cdot x_i$.
- Note that for any x_i, $|G \cdot x_i| \cdot |G_{x_i}| = |G|$.
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We start with $C = \{(g, x) \mid g \cdot x = x\}$:

- $|C| = \sum_{g \in G} |X^g|$.
- $|C| = \sum_{x \in X} |G_x| = \sum_i \left(\sum_{x \in G \cdot x_i} |G_x| \right)$.
- For each orbit $G \cdot x_i$, $|G_x| = |G_{x_i}|$ for all $x \in G \cdot x_i$.
- Note that for any x_i, $|G \cdot x_i| \cdot |G_{x_i}| = |G|$.
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$X =$ all labelings of 6 faces with numbers 1–6.

$$|X| = 6! = 720.$$

$G =$ rotational symmetry group of the cube, $|G| = 24$.

- No non-identity rotation fixes a labelling (you always move at least one face).
- So the action is **fixed-point free**.

Therefore

$$|X/G| = \frac{|X|}{|G|} = \frac{720}{24} = 30.$$

There are **30** distinct dice up to rotation.

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Problem 2: 3-Card Hands from 3 Identical Decks

We consider ordered hands first:

$$H = \{(a_1, a_2, a_3) : 1 \leq a_i \leq 52\}, \quad |H| = 52^3.$$

- $G = S_3$ acts by permuting positions.
- Two hands are "the same" if one can be permuted to get the other.

So we want $|H/S_3|$, the number of orbits of H under S_3.

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Problem 2: Burnside's Lemma for Card Hands

We apply Burnside:

$$|H/S_3| = \frac{1}{|S_3|} \sum_{g \in S_3} |X^g|.$$

- Identity e: fixes all hands, $|X^e| = 52^3$.
- Three transpositions, e.g. (12):
  - Condition: $a_1 = a_2$, but a_3 is free.
  - So $|X^{(12)}| = 52 \cdot 52 \cdot 52 = 52^3$ (same as e).
- Two 3-cycles (123) and (132):
  - Condition: $a_1 = a_2 = a_3$, all three must be equal.
  - So $|X^{(123)}| = 52 \cdot 1 \cdot 1 = 52$.

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Problem 2: Final Counting

Putting everything together:

$$|H/S_3| = \frac{1}{6}\left(52^3 + 3 \cdot 52^2 + 2 \cdot 52\right) = 24804.$$

So there are **24,804** distinct 3-card hands when 3 identical 52-card decks are mixed.

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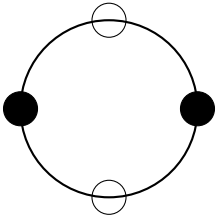
Problem 3: Necklaces with 2 Black, 2 White Beads

We have 4 bead positions equally spaced around a loop.



- Let X be all sequences with 2 black and 2 white beads.
- $G = D_4$ acts by rotating and reflecting the square.
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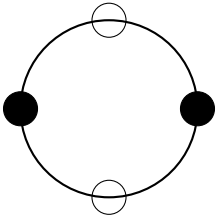

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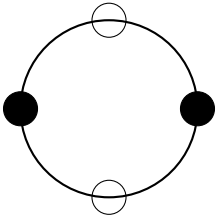

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Problem 3: Counting by Burnside

Without going through every symmetry in detail (left as exercises):

- The identity fixes all 6 sequences of 2 black, 2 white beads.
- (1234) and (1432) have no fixed sequences.
- (12)(34) have 2 fixed sequences. Similarly for (13)(24) and (14)(23).
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Burnside's Lemma gives:

$$|X/D_4| = \frac{1}{8}(6 + 5 \cdot 2) = 2.$$

So there are exactly 2 distinct necklaces with 2 black and 2 white beads.

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- Symmetries can cause **severe overcounting** in combinatorics.
- Group theory provides a **clean language for symmetries**.
- **Group actions** and **Burnside's Lemma** let us count objects up to symmetry.
- We saw applications to *dice labelling*, *card hands*, and *necklaces*.

In CS/AI, similar ideas appear in:

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Thank You